

Amithesh Sathish

Los Angeles, CA · (319) 936-9922 · amitheshsathish@gmail.com
linkedin.com/in/ami03sa · github.com/Ami03sa

Professional Summary

Computer Science graduate student specializing in Artificial Intelligence, Large Language Models (LLMs), and agent-based systems. Experienced in building LLM-powered applications, RAG systems, and multi-step AI agents to solve real-world problems. Interested in developing scalable AI systems, intelligent agents, and data-driven solutions in production environments.

Education

University of Southern California (USC)

M.S. Artificial Intelligence

Los Angeles, CA

Expected Dec 2027

University of Iowa

B.S. Computer Science, Minor in Entrepreneurial Management

Iowa City, IA

June 2025

Honors: Dean's List

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL, Bash

AI/ML: LLMs, VLMs, RAG Systems, Agent Workflows, Prompt Engineering, LoRA Fine-Tuning

Frameworks: PyTorch, Hugging Face, Scikit-learn, XGBoost

Data: Pandas, NumPy, Apache Spark, PostgreSQL, MySQL, GeoTIFF, GeoJSON

Tools: AWS (EC2, Lambda, S3), Docker, Git, Linux, FAISS, Pinecone

Professional Experience

Ursa Space Systems

Ithaca, NY

AI Software Engineer Intern (NY Space Grant)

Apr 2026 – Jul 2026

- Engineered a Vision-Language Model (VLM) designed to scan large-scale GEGD GeoTIFF geospatial imagery for autonomous detection and classification of vessels, ships, and aircraft.
- Systematically analyzed over 300 GB of vendor-supplied GeoTIFF and GeoJSON satellite datasets to assess structural image characteristics, tracking weekly delivery metrics, data payload sizing, and visual clarity.
- Evaluated and validated complex satellite data extraction algorithms to ensure high precision and reporting integrity for defense, finance, and commodity intelligence workflows.

QLT AI

Los Angeles, CA

Founding Engineer (BallTalk)

June 2025 – Present

- Built multi-step LLM agents to generate real-time NBA analytics insights using structured and unstructured data.
- Developed a RAG-based system integrating vector databases and APIs for accurate information retrieval.
- Designed prompt workflows and tool-calling systems enabling autonomous reasoning and data querying.

Warner Music Group

New York, NY

AI Engineering Intern

Aug 2024 – Nov 2024

- Developed LLM-powered analytics tools to generate insights from large-scale engagement data.
- Fine-tuned transformer models on 50k+ labeled samples, improving predictive accuracy by 45%.
- Evaluated model performance and optimized retrieval pipelines to substantially reduce hallucinations.

Warner Music Group

New York, NY

ML Engineering Intern

Nov 2023 – Mar 2024

- Built machine learning models and automated data pipelines for large-scale predictive market analytics.
- Streamlined ETL workflows using Apache Spark, reducing overall data processing turnaround times by 40%.

Projects

Agricultural Intelligence Initiative

University of Iowa

Lead Developer

Oct 2023 – May 2024

- Developed a CNN-based computer vision model for crop disease detection using raw image data to assist rural farmers.
- Built robust, end-to-end machine learning pipelines encompassing data preprocessing, training optimization, and model deployment.

Certifications & Activities

DeepLearning.AI Prompt Engineering, AWS Cloud Practitioner, Databricks Generative AI

Hackathon Winner – University of Iowa

Member – National Society of Black Engineers (NSBE)